Chapter 6

Precipitation titration and gravimetry

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Learning Objectives / 学习要求

Master / 掌握

Master: precipitation–dissolution equilibrium; solubility and conditional solubility product; factors affecting the equilibrium; indicator methods; precipitation conditions; gravimetric factor and mass-percentage calculations.

难溶化合物的沉淀-溶解平衡，溶解度、条件溶度积的含义及影响沉淀平衡的因素；铬酸钾指示剂法、铁铵矾指示剂法和吸附指示剂法指示终点的原理和条件；沉淀重量法中不同类型沉淀的沉淀条件；重量因数（换算因数）及质量百分数的计算方法

Learning Objectives / 学习要求

Familiar with / 熟悉

Become familiar with argentometric/ˌɑːdʒənˈtɒmɪtrɪk/ titration curves; preparation and standardization of standard solutions; and the requirements for precipitate and weighing forms in precipitation gravimetry.

银量法滴定曲线；标准溶液的配制和标定；沉淀重量法中对沉淀形式和称量形式的要求

Learning Objectives / 学习要求

Understand / 理解*（自学）

Understand the applications of precipitation gravimetry; the morphology and formation of precipitates; sources of precipitate impurity and methods for their reduction; and the principles and applications of volatilization gravimetry.

沉淀重量法的应用；沉淀重量法中沉淀的形态和形成过程；造成沉淀不纯的因素及减免方法；挥发重量法的原理及应用

Section 1: Overview

Section 1: Overview

Precipitation titration (沉淀滴定法)

- A titration analysis based on a precipitation reaction
- Among the many reactions that can generate precipitates, very few are really suitable for precipitation titration analysis
- Because the composition of many reactions to generate precipitates is not constant, or the solubility is large, or it is easy to form a supersaturated solution, or the rate of reaching equilibrium is slow, or the co-precipitation phenomenon is serious很多反应生成沉淀的组成不恒定，或溶解度较大，或容易形成过饱和溶液，或达到平衡的速度慢，或共沉淀现象严重等
- At present, the main reaction used is the formation of insoluble silver salt. The precipitation titration method established by this reaction is called argentometry (银量法).

Section 1: Overview

Gravimetric analysis (重量分析法)

- A method of determining the content of the component under test by weighing the mass of the substance称量物质的质量来确定被测组分含量
- This method does not need to be compared with standard samples or reference materials, and there is no error caused by volume vessels
- It is cumbersome, time-consuming, and has a large error in the determination of low content components
- At present, gravimetric analysis is still used for the determination of the content of certain macronutrients such as silicon, sulfur, tungsten, and the moisture, ash, and volatiles of the drug.常量元素如硅、硫、钨的含量和药物的水分、灰分和挥发物等测定仍采用重量分析法

Section 2 Precipitation–Dissolution Equilibrium

Section 2 Precipitation–Dissolution Equilibrium

I. Solubility溶解度

(1) intrinsic solubility，S^0 固有溶解度

Take the 1:1 type sparingly soluble compound MA as an example, the precipitation-dissolution equilibrium of MA in water is

以1:1型难溶化合物MA为例， MA在水中的沉淀—溶解平衡为：

MA（固） ⇋MA（水） ⇋ $M^+ + A^-$

$$S^0 = \frac{a_{\text{MA（水）}}}{a_{\text{MA（固）}}}$$

At a given temperature, the activity of a pure solid $a_{\text{MA（固）}} = 1$， $a_{\text{MA（水）}} = S^0$

Section 2 Precipitation–Dissolution Equilibrium

I. Solubility

(2) Solubility, S溶解度

is the total concentration of MA dissolved in equilibrium. If no other equilibrium relationship exists in the solution, the solubility S of the solid MA shall be the sum of the intrinsic solubility S^0 and the concentration of the crystalline ions M+ or A-:

是在平衡状态下所溶解的MA的总浓度。若溶液中不存在其他平衡关系时，则固体MA的溶解度S应为固有溶解度S^0和构晶离子M^+或A^-的浓度之和：

$$S = S^0 + [M^+] = S^0 + [A^-]$$

Section 2 Precipitation–Dissolution Equilibrium

I. Solubility

(2) Solubility, S溶解度

In general, S0 is very small, and the inherent solubility of AgBr, AgCl, AgIO3, etc. only accounts for the total solubility of 0.1% ~ 1%, and S0 is not easy to measure. Therefore, the inherent solubility is generally negligible. Therefore, the solubility of MA is:

Formula: 一般情况下S0很小，如AgBr、AgCl、AgIO3等的固有溶解度仅占总溶解度的0.1%～1%，且S0也不易测得。因此，固有溶解度一般可忽略不计。所以，MA的溶解度为：

$$S = [M^+] = [A^-]$$

II. Activity product and solubility product

(1) Activity Product活度积

According to the equilibrium relationship of the sparingly soluble compound MA in aqueous solution根据难溶化合物MA在水溶液中的平衡关系

- MA（固）⇋ MA（水）⇋ $M^{+}+A^{-}$

$K=\frac{a_{M^+}\cdot a_{A^-}}{a_{MA（水）}}$，将$S^0$代入得：$a_{M^+}\cdot a_{A^-}=S^0\cdot K=K_{ap}$

- K_{ap} is called the activity product constant活度积常数，abbreviated as solubility product活度积。 K_{ap} is a thermodynamic constant热力学常数that varies with temperature.
- The relationship between activity and concentration is
- $a_{M^+}\cdot a_{A^-}=\gamma_{M^+}\cdot[M^+]\cdot\gamma_{A^-}\cdot[A^-]=K_{ap}$

$$[M^+]\cdot[A^-]=\frac{K_{ap}}{\gamma_{M^+}\cdot\gamma_{A^-}}=K_{sp}$$

II. Activity product and solubility product

(2) Solubility product溶度积

- For the solubility product of sparingly soluble compounds难溶化合物of the M_mA_n type：$K_{sp} = [M^{n+}]^m \cdot [A^{m-}]^n$
- K_{sp} known as the solubility product constant溶度积常数, or solubility product for short溶度积
- K_{sp}与溶液的温度和离子强度有关。当溶液的离子强度较小时，活度系数为1，即$K_{ap}=K_{sp}$，活度积可作为溶度积使用；当溶液的离子强度较大时，$K_{ap}\neq K_{sp}$，此时，应用活度系数（γ）对活度积进行校正

II. Activity product and solubility product

The relational expression between solubility product and solubility is

$$S = [M^+] = [A^-] = \sqrt{K_{sp}} \quad (MA型)$$

$$S = \frac{[M^{n+}]}{m} = \frac{[A^{m-}]}{n} = \sqrt[m+n]{\frac{K_{sp}}{m^m n^n}} \quad (M_mA_n型)$$

III. Conditional solubility product条件溶度积

In addition to the main reaction of the ion to be measured and the precipitating agent to form a precipitate, there are many side reactions in the solution. The strength of the side reactions is described by the side reaction coefficients α_M, α_A For MA type precipitates, in the presence of side reactions:

$$K_{sp} = [M][A] = \frac{[M'][A']}{\alpha_M \cdot \alpha_A} = \frac{K_{sp}'}{\alpha_M \cdot \alpha_A}$$

In the above formula, the charge of M and A is omitted, [M ′] is the total concentration of metal ions, [A ′] is the total concentration of precipitant, K'_{sp} is called conditional solubility product, that is:

$$K_{sp}' = [M'][A'] = K_{sp} \cdot \alpha_M \cdot \alpha_A$$

III. Conditional solubility product

Due to the presence of side reactions, K ′ sp > Ksp, the solubility at this time is:

$$S = [M'] = [A'] = \sqrt{K'_{SP}}$$

M_mA_n型沉淀的溶解度与条件溶度积之间的关系式为：

$$K'_{SP} = K_{sp} \cdot a_M{}^m \cdot a_A{}^n$$

$$S = \sqrt[m+n]{\frac{K_{sp}'}{m^m n^n}}$$

The conditional solubility product K'_{SP} of the same precipitate changes with the precipitation conditions. K'_{SP} can truly and objectively reflect the solubility of the precipitate and its influencing factors.

同一种沉淀的条件溶度积K'_{SP}随沉淀条件的变化而改变。K'_{SP}能真实、客观地反映沉淀的溶解度及其影响因素

Section 2 Precipitation–Dissolution Equilibrium

IV. Factors Affecting the Precipitation-Dissolution Balance

(1) Common ion effect

The common ion effect is the phenomenon that when the precipitation reaction reaches equilibrium, the solubility of the insoluble salt is reduced by increasing the concentration of the appropriate amount of crystalline ions. In gravimetric analysis, excessive precipitating agents are often added to make the precipitation more complete using the isotonic effect.

In general, the excessive amount of precipitant 50% ~ 100% can achieve the intended purpose. If the precipitant is not volatile, the excessive amount of 20% ~ 30% is appropriate. However, if there is too much precipitant, it may cause side reactions such as salt effect, acid effect and coordination effect, which will increase the solubility of the precipitate.

一般情况下，沉淀剂过量50%～100%可达到预期目的，如果沉淀剂不易挥发，则以过量20%～30%为宜。但沉淀剂若过量太多，则又有可能引起盐效应、酸效应及配位效应等副反应，反而使沉淀的溶解度增大

Section 2 Precipitation–Dissolution Equilibrium

IV. Factors Affecting the Precipitation-Dissolution Balance

(2) Acid effect

The acid effect (acid effect) is a phenomenon where the acidity of the solution changes to change the solubility of the insoluble salt. The main reason for this is the effect of H+ in the solution on the dissociation balance of the insoluble salt.

The acid effect has a greater impact on the precipitation of weak acid salts or polyacid salts, precipitations that are themselves weak acids (such as silicic acid), and precipitations formed by many organic precipitants

IV. Factors Affecting the Precipitation-Dissolution Balance

(3) Coordination effect

The coordination effect (complex effect) is a phenomenon that increases the solubility of insoluble salts when there is a coordination agent in the solution that can form a soluble complex with metal ions.

In some precipitation reactions, the synionic effect plays a major role when the precipitating agent is appropriately excessive; the coordination effect plays a major role when the precipitating agent is excessive. For example, Cl- is added to the Ag+ solution to generate a AgCl precipitate, but if an excessive amount of Cl- is added, Cl- can coordinate with AgCl to generate AgCl2- and AgCl32- coordination ions, and the AgCl precipitate is gradually dissolved.

有些沉淀反应，当沉淀剂适当过量时，同离子效应起主要作用；当沉淀剂过量太多时，配位效应起主要作用。例如在Ag+溶液中加入Cl-，生成AgCl沉淀，但若继续加入过量的Cl-，则Cl-能与AgCl配位生成$AgCl_2^-$和$AgCl_3^{2-}$配位离子，而使AgCl沉淀逐渐溶解

IV. Factors Affecting the Precipitation-Dissolution Balance

(4) Salt effect

Salt effect (salt effect) refers to the phenomenon that the solubility of insoluble salts increases with the increase of ionic strength in solution

When the concentration of the strong electrolyte increases, the ionic strength of the solution increases and the ionic activity coefficient decreases

The activity coefficient of high valence ions is greatly affected by the ion strength, so the higher the charge of crystalline ions, the more serious the salt effect.

In order to reduce the impact of the salt effect, the presence of other strong electrolytes should be avoided as much as possible during the precipitation reaction. However, for precipitates with very low solubility, such as $Fe_2O_3 \cdot nH_2O$, the effect of the salt effect is very small and negligible. When the solubility of the precipitate is large, attention must be paid to the effect of the salt effect

为减少盐效应的影响，在进行沉淀反应时应当尽量避免其他强电解质的存在。但是对于溶解度很小的沉淀如$Fe_2O_3 \cdot nH_2O$等，则盐效应的影响非常小，以至可以忽略不计。当沉淀的溶解度较大时，则必须注意盐效应的影响

Section 2 Precipitation–Dissolution Equilibrium

IV. Factors Affecting the Precipitation-Dissolution Balance

V. Other considerations

Effect of 1. temperature

The dissolution reaction is generally an endothermic reaction吸热反应, and the solubility of the precipitate increases with the increase of temperature.

Effect of 2. Solvent

The solubility of most inorganic insoluble salts is greatly affected by the polarity溶剂极性of the solvent. The greater the polarity of the solvent, the greater the solubility of inorganic insoluble salts. Changing the polarity of the solvent can change the solubility of the precipitate.

3. Effect of sediment particle size

4. Hydrolysis

5. Solubilization

Section 3 Precipitation Titration

Section 3 Precipitation Titration

The precipitation titration reaction must meet the following requirements: ① The solubility of the precipitation must be small; ② The precipitation reaction must be completed quickly; ③ The reaction must have a clear metering relationship, and the precipitation composition is constant; ④ There is an appropriate method to indicate the stoichiometry point, and the adsorption phenomenon of the precipitation does not affect the determination of the endpoint.

沉淀滴定的反应必须满足下列几点要求：①沉淀的溶解度要小；②沉淀反应必须迅速完成；③反应要有明确的计量关系，沉淀组成恒定；④有适当的方法指示化学计量点，沉淀的吸附现象不影响终点的确定

Section 3 Precipitation Titration

I. Titration curve of silver method

Taking the NaCl solution of 0.1000mol/L $AgNO_3$ standard solution titration 20.00ml 0.1000mol/L as an example, the titration reaction is:

$$Ag^+ + Cl^- \rightleftharpoons AgCl\downarrow$$

$$K_{sp}=1.8\times10^{-10}\ (pK_{sp}=9.74)$$

(1) Before titration, $[Cl^-]$ = 0.1000mol/L, pCl = 1.00 in solution

Section 3 Precipitation Titration

I. Titration curve of silver method

(2) Calculate pCl based on the remaining [Cl^-] in the solution before the titration starts to the stoichiometric point.滴定开始至化学计量点前，根据溶液中剩余[Cl^-]计算pCl. When $AgNO_3$ solution 19.98ml is added, the remaining Cl- concentration in the solution is:

$$[Cl^-] = \frac{0.1000 \times 0.02}{20.00 + 19.98} = 5.0 \times 10^{-5}\ \text{mol/L}$$

pCl＝4.30　pAg＝5.44

Section 3 Precipitation Titration

I. Titration curve of silver method

(3) When titrated to stoichiometric point, the solution is a saturated solution of AgCl滴定至化学计量点时，溶液为AgCl的饱和溶液

$$[Cl^-] = [Ag^+] = \sqrt{K_{sp}} = \sqrt{1.8 \times 10^{-10}} = 1.3 \times 10^{-5}\ (mol/L)$$

$$pCl = 4.87 \qquad pAg = 4.87$$

Section 3 Precipitation Titration

I. Titration curve of silver method

(4) After the stoichiometric point, Ag^+ is excessive, and pCl is determined by excessive [Ag^+]. When adding $AgNO_3$ solution 20.02ml:

化学计量点后，Ag+过量，pCl由过量的［Ag+］决定

$$[Ag^+] = \frac{0.1000 \times 0.02}{20.00 + 20.02} = 5.0 \times 10^{-5}(\text{mol/L})$$

pAg＝4.30　　pCl＝9.74 - 4.30＝5.44

Section 3 Precipitation Titration

I. Titration curve of silver method

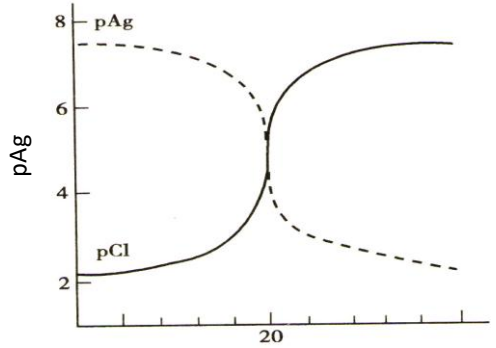
(5) Features: ① When approaching the stoichiometric point, a small amount of Ag+ solution is added, and the concentration of Cl- changes greatly, forming a sudden jump on the titration curve; ② The two curves of pCl and pAg are symmetrical to the stoichiometric point; ③ The size of the sudden jump range is related to the concentration of the solution and the Ksp that generates the precipitation

（5）特点：①接近化学计量点时，加入很少量的Ag^+溶液， Cl^-的浓度发生很大变化，在滴定曲线上形成突跃；②pCl与pAg两条曲线以化学计量点对称；③突跃范围的大小与溶液的浓度和生成沉淀的K_{sp}有关

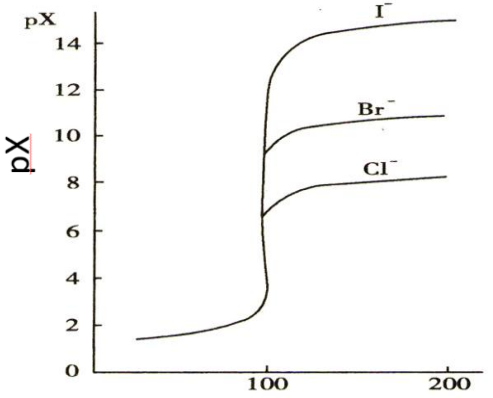
Section 3 Precipitation Titration

I. Titration curve of silver method

(5) Features: ① When approaching the stoichiometric point, a small amount of Ag+ solution is added, and the concentration of Cl- changes greatly, forming a sudden jump on the titration curve; ② The two curves of pCl and pAg are symmetrical to the stoichiometric point; ③ The size of the sudden jump range is related to the concentration of the solution and the Ksp that generates the precipitation



Ag^+ 加入量/ml



Ag^+ 加入百分数/%

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method**银量法**

(1) Potassium chromate indicator method**铬酸钾指示剂法/莫尔法**

1. Principle:

滴定反应：$Ag^+ + Cl^- \rightleftharpoons AgCl\downarrow$（白色）　　$K_{sp}=1.8\times10^{-10}$

终点反应：$2Ag^+ + CrO_4^{2-} \rightleftharpoons Ag_2CrO_4\downarrow$（砖红色）　　$K_{sp}=1.2\times10^{-12}$

莫尔法是以创立该方法的德国化学家 Karl Friedrich Mohr（卡尔·弗里德里希·莫尔，1806–1879）的姓氏命名。

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method

(1) Potassium chromate indicator method

2. titration conditions

(1) Indicator dosage: The dosage of the indicator CrO_4^{2-} directly affects the accuracy of the Mohr method. The concentration of CrO_4^{2-} is too high, not only the endpoint is advanced, but also the yellow color of CrO_4^{2-} itself will affect the endpoint observation; the concentration of CrO_4^{2-} is too low, and the endpoint is delayed; when actually titrating, generally add 5% potassium chromate indicator 1 ~ 2ml in a solution with a total volume of 50 ~ 100ml, at this time, $[CrO_4^{2-}]$ is 2.6×10^{-3} ~ 5.2×10^{-3}mol/L

（1） 指示剂的用量：指示剂CrO_4^{2-}的用量直接影响莫尔法（Mohr法）的准确度。CrO_4^{2-}的浓度过高，不仅终点提前，而且CrO_4^{2-}本身的黄色也会影响终点观察；CrO_4^{2-}的浓度过低，终点滞后；实际滴定时，一般是在总体积为50～100ml的溶液中加入5%铬酸钾指示剂1～2ml，此时，$[CrO_4^{2-}]$为2.6×10^{-3} ～5.2×10^{-3}mol/L

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method
(1) Potassium chromate indicator method
2. titration conditions
(1) Indicator dosage:
The total consumption of the AgNO3 standard solution should be appropriate during the titration process. If the volume consumption of the standard solution is too small, or the concentration of the standard solution is too low, the relative error of the test results will increase due to the excess of the endpoint. To do this, “blank correction” of the indicator must be done
在滴定过程中，$AgNO_3$标准溶液的总消耗量应适当。若标准溶液体积消耗太小，或标准溶液浓度过低，都会因为终点的过量使测定结果的相对误差增大。为此，须做指示剂的“空白校正”

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method

(1) Potassium chromate indicator method

2. titration conditions

(2) Solution acidity: The acidity of the solution is too high, and CrO_4^{2-} generates a weak electrolyte H_2CrO_4 or $Cr_2O_7^{2-}$, which causes the concentration of CrO_4^{2-} to decrease, and the precipitation of Ag_2CrO_4 is delayed, or even not precipitated; if the acidity of the solution is too low, Ag_2O precipitation is generated. Experiments have shown that the Mohr method should be performed in a neutral or weakly alkaline medium of pH 6.5 to 10.5

溶液的酸度：溶液的酸度过高，CrO_4^{2-}生成弱电解质H_2CrO_4或$Cr_2O_7^{2-}$，致使CrO_4^{2-}浓度降低，Ag_2CrO_4沉淀推迟，甚至不产生沉淀；溶液的酸度过低，则生成Ag_2O沉淀。实验证明，莫尔法应在pH 6.5～10.5的中性或弱碱性介质中进行

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method

(1) Potassium chromate indicator method

2. titration conditions

(3) Titration should be vigorously shaken: Vigorously shaking can release Cl- or Br- adsorbed by AgCl or AgBr precipitation, preventing the focus from being advanced

（3）滴定时应剧烈振摇：剧烈振摇可释放出被AgCl或AgBr沉淀吸附的Cl^-或Br^-，防止重点提前

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method

(1) Potassium chromate indicator method

2. titration conditions

(4) Elimination of interference: The anions that generate precipitates with Ag+ such as PO_4^{3-}、AsO_4^{3-}、SO_3^{2-}、S^{2-}、CO_3^{2-}、CrO_4^{2-}, the cations that generate precipitates with CrO42- such as Ba2+, Pb2+, etc., a large number of colored ions such as Cu^{2+}、Co^{2+}、Ni^{2+}, etc., ions that are prone to hydrolysis in neutral or weakly alkaline solutions such as Fe^{3+}、Al^{3+}、Bi^{3+}、Sn^{4+}, etc. interfere with the determination, and should be separated in advance

（4）干扰的消除：与Ag^+生成沉淀的阴离子如PO_4^{3-}、AsO_4^{3-}、SO_3^{2-}、S^{2-}、CO_3^{2-}和CrO_4^{2-}等，与CrO_4^{2-}生成沉淀的阳离子如Ba^{2+}、Pb^{2+}等，大量Cu^{2+}、Co^{2+}、Ni^{2+}等有色离子，在中性或弱碱性溶液中易发生水解反应的离子如Fe^{3+}、Al^{3+}、Bi^{3+}和Sn^{4+}等均干扰测定，应预先分离

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method

(1) Potassium chromate indicator method

3. Scope of application应用范围

The Mohr method is mainly used for the determination of Cl^-, Br^- and CN^-, and is not suitable for the determination of I^- and SCN^-, and is not suitable for the direct titration of Ag+ with NaCl standard solution. If Ag^+ is determined by the Mohr method, the back titration method must be used

Mohr法主要用于测定Cl^-、Br^-和CN^-，不适用于测定I^-和 SCN^-，不适用于以NaCl标准溶液直接滴定Ag^+。如用Mohr法测定Ag^+，必须采用返滴定法

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method

(2) Ferric ammonium sulfate indicator method

铁铵矾指示剂法/福尔哈德Volhard法

1. direct titration:

滴定反应：$Ag^+ + SCN^- \rightleftharpoons AgSCN\downarrow$（白色） $K_{sp(AgSCN)}=1.1\times10^{-12}$

终点反应：$Fe^{3+} + SCN^- \rightleftharpoons [Fe(SCN)]^{2+}$（红色）$K=200$

滴定应在0.1～1mol/L HNO_3介质中进行。酸度过低，Fe^{3+}发生水解，生成$[Fe(OH)]^{2+}$等一系列深色配合物，影响终点观察

在终点恰好能观察到$[Fe(OH)]^{2+}$明显的红色，所需的$[Fe(OH)]^{2+}$最低浓度为6×10^{-6}mol/L

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method
(2) Ferric ammonium sulfate indicator method
(2) Back titration:返滴定法

滴定反应：Ag^+（定量，过量）+ X^- ⇌ AgX↓

Ag^+（剩余量）+ SCN^- ⇌ AgSCN↓（白色）

终点反应：$SCN^- + Fe^{3+}$ ⇌ $[Fe(SCN)]^{2+}$（红色）

Titration should be performed in 0.1 to 1mol/L HNO_3 medium

滴定应在0.1～1mol/L HNO_3介质中进行

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method
(2) Ferric ammonium sulfate indicator method
(2) Back titration method: In order to avoid the occurrence of precipitation conversion reaction, one of the following measures should be taken: ① After precipitation and filtration of the generated AgCl, titrate the filtrate with NH4SCN standard solution; ② Before titration with NH4SCN standard solution, add 1 ~ 2ml nitrobenzene or 1, 2- dichloroethane to the precipitation to generate AgCl; ③ Increase the concentration of Fe3+ to reduce the concentration of SCN- at the endpoint, thereby reducing the titration error

（2）返滴定法：为避免沉淀转化反应的发生，应采取下列措施之一：①将已生成的AgCl沉淀滤去后，再用NH4SCN标准溶液滴定滤液；②用NH4SCN标准溶液滴定前，在生成AgCl的沉淀中加入1～2ml硝基苯或1，2-二氯乙烷；③提高Fe3+的浓度，以减小终点时SCN-的浓度，从而减小滴定误差

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method

(3) Adsorption indicator method

吸附指示剂法/法扬斯法Fajans法

1. Principle

Adsorption Indicator is a class of colored organic dyes that, when adsorbed by charged precipitated gel particles, cause a change in color due to a change in their structure to indicate the titration endpoint

吸附指示剂是一类有色的有机染料，当它被带电的沉淀胶粒吸附时，因其结构改变而导致颜色改变，以此指示滴定终点

终点前 Cl^-过量　　$AgCl\cdot Cl^- + FIn^-$（黄绿色）

终点后 Ag^+过量　　$AgCl\cdot Ag^+ + FIn^- \rightleftharpoons AgCl\cdot Ag^+\cdot FIn^-$（粉红色）

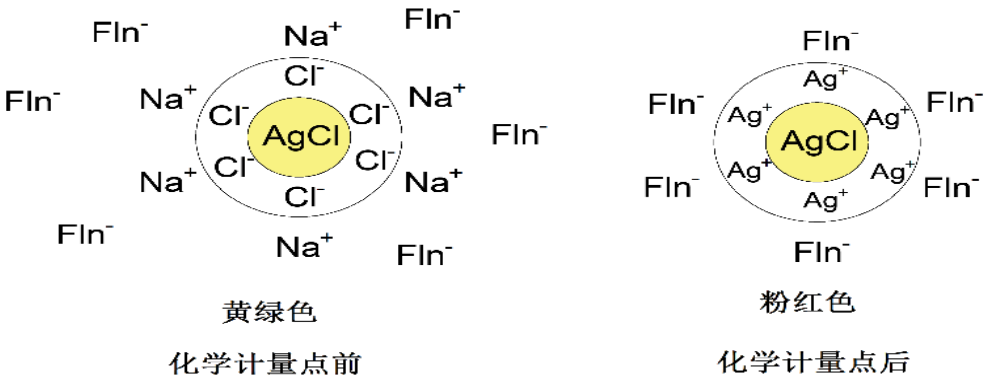
Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method

(3) Adsorption indicator method

1. Principle

沉淀表面电荷变化，不吸附->吸附指示剂



荧光黄吸附指示剂变色原理示意图

Schematic diagram of the discoloration mechanism of fluorescent yellow adsorption indicator

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method
(3) Adsorption indicator method
2. titration conditions
(1) The specific surface area of the precipitation should be as large as possible The acidity of the
(2) solution should be conducive to the presence of the indicator chromogenic form
(3) The adsorption capacity of colloidal particles to the indicator should be slightly less than the adsorption capacity of the tested ions
（1）沉淀的比表面积要尽可能的大
（2）溶液的酸度应有利于指示剂显色型体的存在
（3）胶体颗粒对指示剂的吸附能力应略小于对被测离子的吸附能力

Section 3 Precipitation Titration

II. Titration endpoint indication method by silver method
(3) Adsorption indicator method
2. titration conditions
(4) Titration should avoid exposure to strong light
(5) Solution concentration should not be too dilute
（4）滴定应避免强光照射
（5）溶液浓度不能太稀

3. Range of application The adsorption indicator method can be used for the determination of Cl^-、Br^-、I^-、SCN^-、SO_4^{2-} and Ag^+ plasma

3.应用范围　吸附指示剂法可用于Cl^-、Br^-、I^-、SCN^-、SO_4^{2-}和Ag^+等离子的测定

Section 3 Precipitation Titration

III. Application of precipitation titration method

(1) Preparation and calibration of standard solution**标准溶液配制和标定**

1. reference substance**基准物质**

(1) Silver nitrate reference substance硝酸银基准物质

(2) Sodium chloride reference substance氯化钠基准物质

2. standard solution**标准溶液**

(1) Silver nitrate standard solution硝酸银标准溶液

(2) standard solution of ammonium thiocyanate (or potassium thiocyanate) 硫氰酸铵（或硫氰酸钾）标准溶液

Section 3 Precipitation Titration

III. Application of precipitation titration method

(2) Application Examples

Assay of example ammonium chloride氯化铵的含量测定

The Pharmacopoeia of the People's Republic of China (2020 annual edition) uses the adsorption indicator method to determine the content of ammonium chloride in ammonium chloride tablets

Take about 0.12g of this product, precisely weigh, add water 50ml to dissolve, then add dextrin solution (1→50) 5ml, 8 drops of fluorescent yellow indicator solution and calcium carbonate 0.10g, shake well, and titrate with silver nitrate titration solution (0.1mol/L). NH4Cl equivalent to 5.349mg per 1ml silver nitrate titration solution (0.1mol/L)

取本品约0.12g，精密称定，加水50ml使溶解，再加糊精溶液（1→50）5ml、荧光黄指示液8滴与碳酸钙0.10g，摇匀，用硝酸银滴定液（0.1mol/L）滴定。每1ml硝酸银滴定液（0.1mol/L）相当于5.349mg的NH_4Cl

Section 3 Precipitation Titration

III. Application of precipitation titration method

(2) Application Examples

Determination of the content of dimercaptosuccinic acid (ammonium iron indicator method)二巯丁二酸的含量测定（铁铵矾指示剂法）

取本品约0.05g，精密称定，置具塞锥形瓶中，加无水乙醇30ml使溶解，加稀硝酸2ml，精密加硝酸银滴定液（0.1mol/L）25ml，强力振摇，置水浴中加热2～3分钟，放冷，滤过，用水洗涤锥形瓶与沉淀至洗液无银离子反应，合并滤液与洗液，加硝酸2ml与硫酸铁铵指示液2ml，用硫氰酸铵滴定液（0.1mol/L）滴定，并将滴定的结果用空白试验校正。每1ml硝酸银滴定液（0.1mol/L）相当于4.556mg的$C_4H_6O_4S_2$

Section 4 Gravimetric Analysis

Section 4 Gravimetric Analysis

- In gravimetric analysis, the measured component is first separated from the sample, converted to a certain weighing form, and then the content of the component is determined by weighing method.
- 在重量分析中，一般先使被测组分从试样中分离出来，转化为一定的称量形式，然后用称量的方法测定该成分的含量
- Mainly divided into precipitation method沉淀重量法, volatilization method沉淀重量法and extraction method萃取法

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(1) Precipitation morphology and formation of precipitation

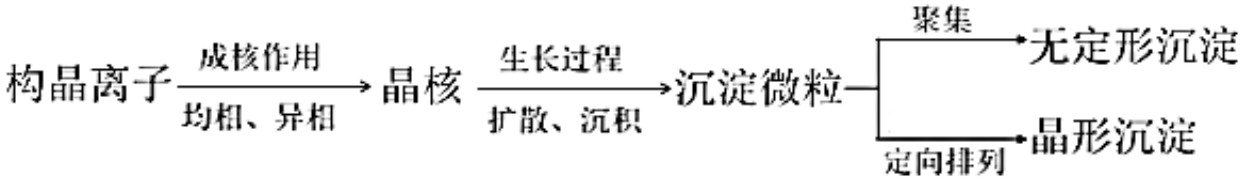
- **沉淀形态与沉淀的形成**
- 1. Form of precipitation
- Crystalline precipitate晶形沉淀: particle diameter is generally 0.1 ~ 1μm
- Amorphous precipitate无定形沉淀 or amorphous precipitate非晶形沉淀: Particle diameter is generally less than 0.02μm
- Curdy precipitate凝乳状沉淀: particle diameter between

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(1) Precipitation morphology and formation of precipitation

2. Formation process of precipitate



(1) Crystal nucleation: Crystal nucleation is generally divided into homogeneous nucleation and heterogeneous nucleation

(2) Nucleus growth: formation of precipitated particles

（1）晶核的形成：晶核的形成一般分为均相成核和异相成核

（2）晶核长大：形成沉淀颗粒

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

Formation process of 2. precipitate

(3) Formation of precipitates:

The rate at which precipitated particles aggregate into larger aggregates is called the aggregation velocity沉淀颗粒聚集成更大聚集体的速度称为聚集速度

The aggregation rate is mainly determined by the supersaturation of the solution聚集速度主要由溶液的过饱和度决定

Q加入滴定剂时溶质的瞬间浓度，S沉淀的溶解度，Q-S过饱和度，K为与沉淀性质、温度和介质等有关的常数

$$v = \frac{K\ (Q - S)}{S}$$

$$v = \frac{K\ (Q - S)}{S}$$

I. Precipitation gravimetry

Formation process of 2. precipitate

(3) Formation of precipitates:

The greater the relative supersaturation, the greater the aggregation velocity, forming an amorphous precipitate; the smaller the relative supersaturation, the smaller the aggregation velocity, which may form a crystalline precipitate

The relative supersaturation decreases with the increase of the solubility of the precipitate S, so the precipitate with large solubility, small aggregation speed, and easy to form crystalline precipitate; otherwise, it is easy to form amorphous precipitate

- 相对过饱和度越大，聚集速度越大，形成无定形沉淀；相对过饱和度越小，聚集速度越小，可能形成晶形沉淀
- 相对过饱和度随沉淀的溶解度*S*的增大而减小，所以溶解度较大的沉淀，聚集速度较小，易生成晶形沉淀；反之则易形成无定形沉淀

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(2) Purity of precipitation and its influencing factors

1. co-precipitation共沉淀

is one of the main sources of gravimetric errors

(1) Surface adsorption: ① Preferential adsorption of ions with compounds with low solubility generated from crystalline ions; ② Ions with the same concentration, the more charged ions, the easier it is to be adsorbed; ③ Ions with the same charge, the greater the concentration, the easier it is to be adsorbed. At the same time, the larger the total surface area of the precipitation, the lower the temperature, and the more the adsorption impurities

（1）表面吸附：①与构晶离子生成溶解度小的化合物的离子优先吸附；②浓度相同的离子，带电荷越多的离子，越易被吸附；③电荷相同的离子，浓度越大，越易被吸附。同时沉淀的总表面积越大，温度越低，吸附杂质量越多

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(2) Purity of precipitation and its influencing factors

1. co-precipitation

(2) Formation of mixed crystals or solid solutions: Mixed crystalline co-precipitation (mixed crystal coprecipitation) is when the adsorbed impurities have the same lattice, charge or ion radius as the precipitation, and the impurity ions can enter the lattice arrangement to cause coprecipitation

（2）形成混晶或固溶体：混晶共沉淀（mixed crystal coprecipitation）是如果被吸附的杂质与沉淀具有相同的晶格、相同的电荷或离子半径，杂质离子可进入晶格排列引起的共沉淀

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(2) Purity of precipitation and its influencing factors

1. co-precipitation

(3) Embedding or adsorption: Embedded co-precipitation (occlusion coprecipitation) is due to the rapid formation of precipitation, the impurities adsorbed on the surface of the precipitation or the mother liquor do not leave in time, covered by the subsequent growth of the precipitation, coated in the precipitation caused by the co-precipitation

（3）包埋或吸留：包埋共沉淀（occlusion coprecipitation）是由于沉淀形成速度快，吸附在沉淀表面的杂质或母液来不及离开，被随后长大的沉淀所覆盖，包藏在沉淀内部引起的共沉淀

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(2) Purity of precipitation and its influencing factors

2. post-precipitation后沉淀

Due to the adsorption effect of the precipitated surface, that is, after precipitation, the precipitated components could not be precipitated in the solution, but also gradually deposited on the precipitated surface. The way to reduce or eliminate post-precipitation is to shorten the time between precipitation and mother liquor co-location

由于沉淀表面的吸附作用引起的，即在沉淀析出后，溶液中本来不能析出沉淀的组分，也在沉淀表面逐渐沉积出来的现象。减少或消除后沉淀的方法是缩短沉淀和母液共置的时间

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(3) Selection of sedimentation conditions

1. Precipitation conditions for crystalline precipitation**晶型沉淀的沉淀条件**

(1) Precipitation in an appropriate dilute solution

(2) Precipitation in hot solution

(3) Slowly add the precipitant under constant stirring

（1）在适当的稀溶液中进行沉淀

（2）在热溶液中进行沉淀

（3）在不断搅拌下，慢慢加入沉淀剂

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(3) Selection of sedimentation conditions

1. Precipitation conditions for crystalline precipitation

(4) Aging 陈化: The process of placing the precipitate with the mother liquor The results of aging are: ① small particles continue to dissolve and large particles continue to grow; ② impurities adsorbed, absorbed or encapsulated in small grains re-enter the solution to make the precipitation more pure; ③ incomplete grains are converted into more complete grains

（4）陈化（aging）：将沉淀与母液一起放置的过程

陈化的结果是：①小颗粒不断溶解，大颗粒不断长大；②吸附、吸留或包藏在小晶粒内部的杂质重新进入溶液，使沉淀更加纯净；③不完整的晶粒转化为更完整的晶粒

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(3) Selection of sedimentation conditions

1. Precipitation conditions for crystalline precipitation

(5) Uniform precipitation: First, control certain conditions so that the added precipitant does not immediately generate precipitate with the tested ions, but slowly and uniformly produces the precipitate from the solution through a chemical reaction, so that the precipitate is slowly and uniformly precipitated throughout the solution.均匀沉淀：先控制一定的条件，使加入的沉淀剂不能立刻与被测离子生成沉淀，而是通过一种化学反应使沉淀剂从溶液中缓慢地、均匀地产生出来，从而使沉淀在整个溶液中缓慢地、均匀地析出

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(3) Selection of sedimentation conditions

2. Precipitation conditions for amorphous precipitation

无定形沉淀的沉淀条件

(1) Precipitation in a thicker hot solution

(2) Add a large amount of electrolyte

(3) Continuous stirring to appropriately speed up the addition of the precipitant

(4) No need to age

（1）在较浓的热溶液中进行沉淀

（2）加入大量电解质

（3）不断搅拌，适当加快沉淀剂的加入速度

（4）不必陈化

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

*了解(4) Filtration and drying of precipitates

1. Filtration of precipitation

According to the nature of the precipitation, select quantitative filters with different densities: ① loose filter paper is used for amorphous precipitation to increase the filtration speed; ② tighter filter paper can be used for crystalline precipitation; ③ Tight filter paper is used for precipitation of fine particles to prevent precipitation from passing through the filter paper

The filtration is carried out by the "pouring method". If the solubility of the precipitate does not change much with the increase of temperature, it can be filtered while it is hot, the effect is better

根据沉淀的性质，选择疏密程度不同的定量滤纸：①无定形沉淀选用疏松滤纸，以增加滤过速度；②晶形沉淀可用较紧密滤纸；③细小颗粒的沉淀应用紧密滤纸，以防沉淀穿过滤纸

滤过时均采用“倾泻法”。若沉淀的溶解度随温度升高变化不大时，可采用趁热滤过，效果更好

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

*了解(4) Filtration and drying of precipitates

2. Washing of precipitates

Suitable washing solutions: ① Precipitates with low solubility and not easy to form colloids, which can be washed with distilled water; ② Crystal precipitates with high solubility can be washed with a dilute precipitant, or can be washed with a precipitated saturated solution; ③ For amorphous precipitates that are easily soluble, wash with a dilute solution of a volatile electrolyte; ④ When the solubility of the precipitate does not change much with increasing temperature, wash with a hot solution

合适的洗涤液：①溶解度小而不易形成胶体的沉淀，可用蒸馏水洗涤；②溶解度大的晶形沉淀可用稀沉淀剂洗涤，也可用沉淀饱和溶液洗涤；③对易胶溶的无定形沉淀，应用易挥发的电解质的稀溶液洗涤；④沉淀的溶解度随温度升高变化不大时，可用热溶液洗涤

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

*了解(4) Filtration and drying of precipitates

3. Drying, burning and constant weight of precipitation

Drying is the weighing form of heating 40 ~ 60 at 110 ~ 120 ℃ for minutes to remove the moisture and volatile substances in the precipitate to obtain the precipitate. Burning is above 800 ℃, completely removing moisture and volatile substances, and decomposing the precipitate into a constant weighing form

3.沉淀的干燥、灼烧和恒重

干燥是在110～120℃下加热40～60分钟，以除去沉淀中的水分和挥发性物质得到沉淀的称量形式。灼烧是在800℃以上，彻底去除水分和挥发性物质，并使沉淀分解为组成恒定的称量形式

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(5) Precipitation form and weighing form**沉淀形式与称量形式**

$SO_4^{2-} + Ba^{2+} \rightleftharpoons BaSO_4\downarrow \xrightarrow{\text{滤过、洗涤}} \xrightarrow{\text{灼烧}} BaSO_4$

$Ca^{2+} + C_2O_4^{2-} \rightleftharpoons CaC_2O_4\downarrow \xrightarrow{\text{滤过、洗涤}} \xrightarrow{\text{灼烧}} CaO$

沉淀形式　　称量形式

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(5) Precipitation form and weighing form

1. Requirements for precipitate forms ① The solubility of the precipitate is small; ② The precipitate must be pure; ③ The precipitate should be easy to filter and wash; ④ The precipitate form is easy to convert into a weighing form with a fixed composition

1.对沉淀形式的要求 ①沉淀的溶解度要小；②沉淀必须纯净；③沉淀应易于过滤和洗涤；④沉淀形式易于转化为具有固定组成的称量形式

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(5) Precipitation form and weighing form

2. Requirements for the form of weighing ① The form of weighing must be fixed and have a definite chemical formula to calculate the result; ② The form of weighing must have sufficient chemical stability, not be affected by factors such as moisture in the air, CO2 and O2, and should not decompose or deteriorate; ③ The form of weighing should have the largest possible molar mass, reduce weighing errors, and improve the accuracy of the analysis results

2.对称量形式的要求 ①称量形式必须组成固定，有确定的化学式才能进行结果计算；②称量形式必须有足够的化学稳定性，不受空气中水分、CO_2和O_2等因素的影响而发生变化，本身也不应分解或变质；③称量形式应具有尽可能大的摩尔质量，减少称量误差，提高分析结果的准确度

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(6) Calculation of gravimetric analysis

Gravimetric factor重量因数or conversion factor换算因数is the ratio of the relative molecular weight of the tested component to the relative molecular weight of the weighed form, commonly expressed as F

被测组分的相对分子质量与称量形式的相对分子质量之比，常用*F*表示

$$\text{换算因数（}F\text{）} = \frac{a \times \text{被测组分的相对分子质量}}{b \times \text{称量形式的相对分子质量}}$$

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(6) Calculation of gravimetric analysis

The mass fraction of the component under test can be obtained from the mass m in the weighed form, the mass m ′ of the specimen, and the conversion factor F由称得的称量形式的质量m、试样的质量m'及换算因数F即可求得被测组分的质量分数

$$w(\%) = \frac{m' \cdot F}{m} \times 100\%$$

Section 4 Gravimetric Analysis

I. Precipitation gravimetry

(6) Calculation of gravimetric analysis

As an example, the potassium bicarbonate oxalate sample 0.5172g was taken, dissolved, and precipitated with Ca^{2+}. After burning, CaO weighed 0.2265g, and the mass fraction of $KHC_2O_4 \cdot H_2C_2O_4 \cdot 2H_2O$ in the sample was calculated.

Solution: $KHC_2O_4 \cdot H_2C_2O_4 \cdot 2H_2O \sim 2CaC_2O_4 \sim 2CaO$

$$F = \frac{M_{KHC_2O_4 \cdot H_2C_2O_4 \cdot 2H_2O}}{2M_{CaO}} = \frac{254.2}{2 \times 56.08} = 2.266$$

$$w(\%) = \frac{m' \cdot F}{m} \times 100\% = \frac{0.2265 \times 2.266}{0.5172} \times 100\% = 99.24\%$$

Section 4 Gravimetric Analysis

*以下内容不要求

2. Volatile gravimetry

(1) Direct method

The direct volatilization method is a method that uses heating and other methods to escape the volatile components in the sample, absorb them all with a suitable absorbent, and calculate the content of this component according to the increase in the weight of the absorbent.

(2) Indirect method

The indirect volatilization method is to use heating and other methods to escape the volatile components in the sample, weigh the residue, and calculate the content of the volatile components according to the difference in the mass of the sample before and after volatilization.

Section 4 Gravimetric Analysis

2. Volatile gravimetry

(2) Indirect method

1. Presence of water in the sample

(1) Hygroscopic water (wet storage water, absorbent water)

(2) Embedded water

(3) Inhaled water

(4) Crystalline Water

(5) Composition water

Section 4 Gravimetric Analysis

2. Volatile gravimetry

(2) Indirect method

2. Common methods for determining loss on drying

(1) Heating and drying at atmospheric pressure: suitable for specimens that are stable in properties such as heat, not easily decomposed, deteriorated, oxidized or volatilized

(2) Decompression Heated Drying: Suitable for drying specimens that are prone to deterioration, have a low melting point, or are less volatile in moisture at high temperatures

(3) Desiccant drying: suitable for substances that can sublimate or be unstable under heat and are prone to deterioration

Section 4 Gravimetric Analysis

III. Application of Gravimetric Analysis

Determination of the content of barium sulfate (type I)
The "Pharmacopoeia of the People's Republic of China" (2020 annual edition) uses the precipitation gravimetric method to determine the content of barium sulfate (type I) Description: The weighing form is barium chromate, and the conversion factor is:

$$F = \frac{M_{BaSO_4}}{M_{BaCrO_4}} = \frac{233.39}{253.32} = 0.9213$$

Section 4 Gravimetric Analysis

III. Application of Gravimetric Analysis

Example volatile ether leachate assay

Take the test article (through the fourth sieve) 2 ~ 5g, precisely weigh it, put it in a phosphorus pentoxide dryer to dry 12 for an hour, put it in a Soxhlet extractor, add an appropriate amount of ether, unless otherwise specified, heat and reflux 8 for an hour, take the ether solution, dry it to a constant weight evaporation dish, place it, volatilize the ether, and place the residue in a phosphorus pentoxide dryer to dry 18 for an hour, precisely weigh it, slowly heat it to 105 ℃, and dry it to constant weight at 105 ℃. Its weight loss is the weight of the volatile ether extract

Volatile ether leachate determination under the four general rules 2201 of the Pharmacopoeia of the People's Republic of China (2020 annual edition) uses volatile weight method

Summary of Section

1. Overview: definition of precipitation titration and gravimetry.

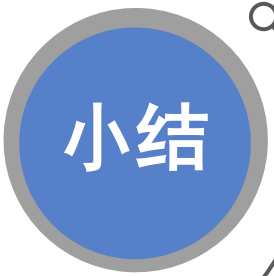
2. Precipitation–dissolution equilibrium: solubility, activity product, solubility product, conditional solubility product, and factors affecting the equilibrium.

3. Precipitation titration: basic principles, Mohr method, Volhard method, adsorption indicator method, standard solutions, and applications.

4. Gravimetric analysis: precipitate formation and purity, precipitation conditions, filtration and drying, weighing form, gravimetric calculations, and volatilization gravimetry.

本章小结

人民卫生出版社
PEOPLE'S MEDICAL PUBLISHING HOUSE



1.概述

沉淀滴定和重量分析法的定义

2.难溶化合物的沉淀-溶解平衡

溶解度、活度积和溶度积、条件溶度积、影响沉淀平衡的因素

3.沉淀滴定法

基本原理、三种滴定终点指示方法、标准溶液和基准物质、应用

4.重量分析法

沉淀重量法中沉淀形态和形成、沉淀纯度和影响因素、沉淀条件选择、沉淀的过滤和干燥、沉淀形式和称量形式、结果计算；挥发重量法中的直接法和间接法；重量法应用

Thank you for your time.